# FEBRIS OPERATING INSTRUCTIONS

Thank you for choosing the Febris sensor from Sentinum. Please read the following operating instructions carefully to prevent damage to the sensor, yourself and the environment.

# Table of contents

Sentinum

# 1. WARNING AND SAFETY INSTRUCTIONS

Warnings and important information about potential hazards or damage

Important information required for smooth operation of the devices

Please note:

- Observe the safety instructions and installation instructions in the manual and the installation list.
- Ensure that the installation environment complies with the prescribed application area guidelines. Observe the temperature and other limit values at all times.
- The device may only be used in the areas specified in the technical specifications.
- The device may only be used for the intended purpose described.
- Safety and functionality can no longer be guaranteed if the device is modified or extended.
- The purpose of the sensor is to monitor indoor air quality.
- **Do not use the sensor in safety-critical applications!**
- **The sensor is expressly <u>not</u> suitable for monitoring carbon monoxide CO!**
- **The sensor is not a smoke detector!**
- The sensor must not be mounted on ceilings or floors
- Operation of the sensor is only permitted up to a maximum of 2000 meters above sea level.
- Due to human exposure regulations, a minimum distance of 20 cm must be maintained between the device and persons.

If the device is installed **incorrectly**:

- It may not function properly.
- It could be permanently damaged.
- It could pose a risk of injury.

Please note:

- Improper handling, e.g. improper mechanical stress, such as dropping the appliance, can result in damage.
- If battery cells other than those recommended are used, performance and product safety may be negatively affected.

Please note $CO_2$ sensor:

- The CO2 measuring module is particularly sensitive to mechanical stress. A fall, even from a low height, as well as shocks and impacts can damage the device. This can lead to incorrect measurements.

## 2. FURTHER DOCUMENTATION

Please note the information and limit values in the technical data sheet.

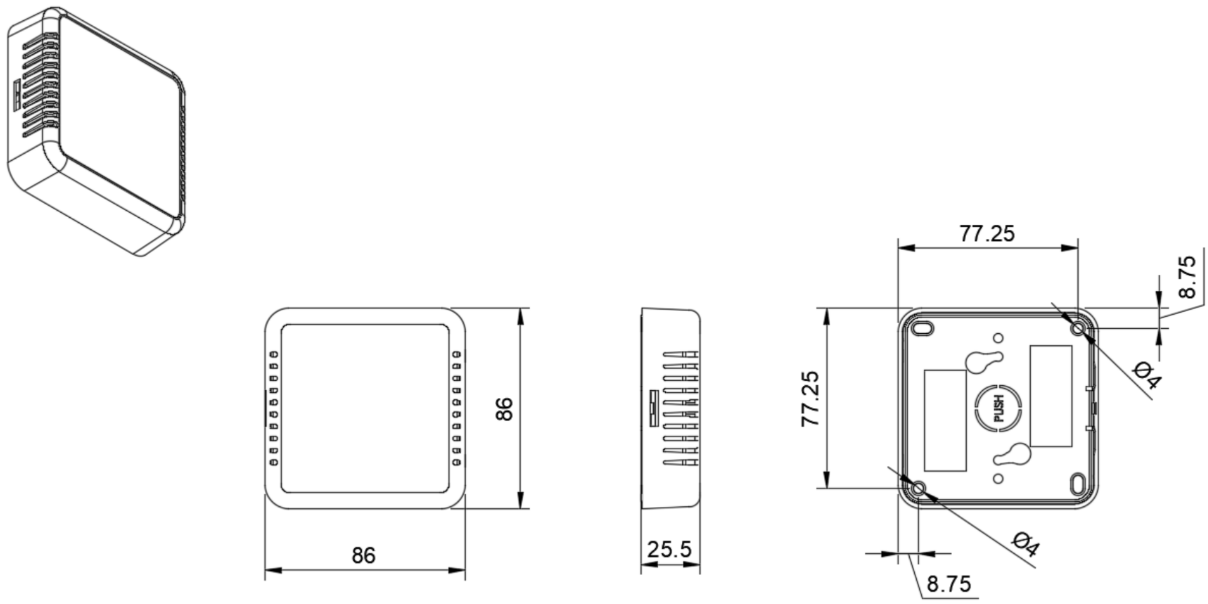
The Sentiface sensor-specific documents can be found in the NFC and downlink description document, the Senticom and Sentivisor tables can be found in the generic NFC and downlink documentation.

The option for configuring sensor communication can be found in the respective generic LoRaWAN® or Mioty® documentation, depending on the version.

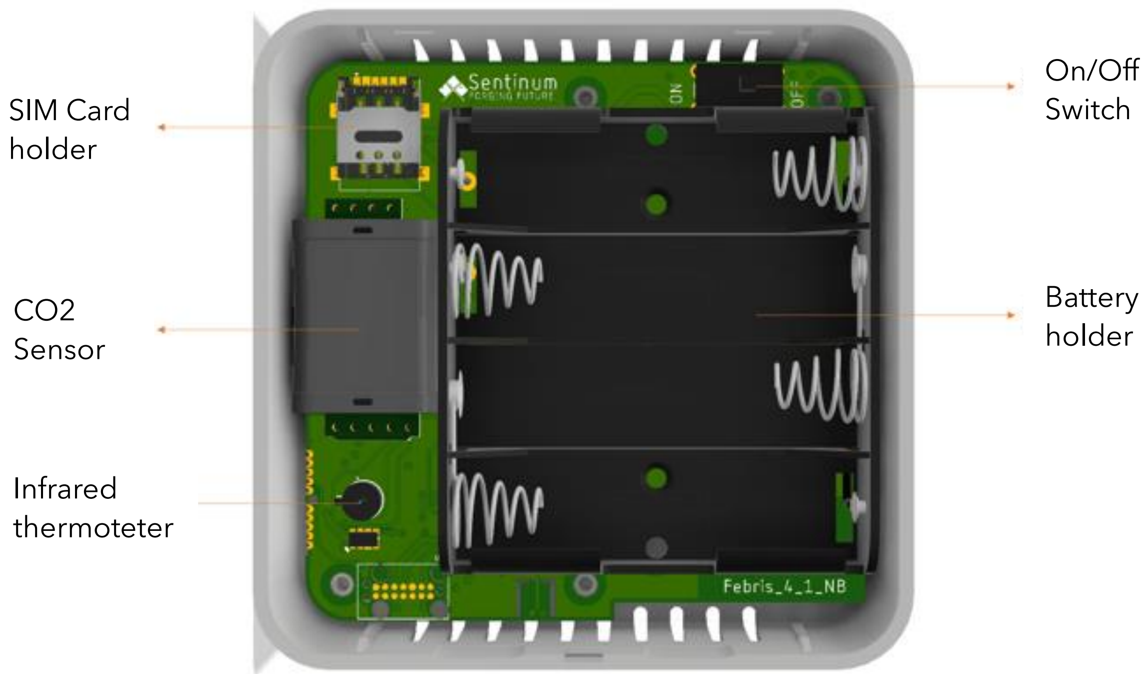
Further generic documentation can be found at https://docs.sentinum.de/en/important-cross-product-documentation-for-sensors.

# 3. DIMENSIONS AND COMPONENTS OF THE SENSOR

## 3.1 TECHNICAL DRAWING




## 3.2 IMPORTANT COMPONENTS



# 4. SCOPE OF DELIVERY AND VERSIONS

The scope of delivery includes:

- 4x batteries for BAT version (or former power supply unit for NET)
- Febris sensor in one of the following versions

| | | |
|---|---|---|
| FEBR-LOEU/MIOTY-CO2-BAT | LoRa ® | THP, CO2 |
| FEBR-LOEU/MIOTY-VOC-BAT | LoRa ® | THP, VOC |
| FEBR-NB-CO2-BAT | NB-IoT, LTE-CATM1 | THP, CO2 |
| FEBR-NB-VOC-BAT | NB-IoT, LTE-CATM2 | THP, VOC |
| FEBR-LOEU/MIOTY-CO2-NET | LoRa ® | THP, CO2 |
| FEBR-LOEU/MIOTY-VOC-NET | LoRa ® | THP, VOC |
| FEBR-NB-CO2-NET | NB-IoT, LTE-CATM1 | THP, CO2 |
| FEBR-NB-VOC-NET | NB-IoT, LTE-CATM2 | THP, VOC |

FEBR – LOEU – CO2 – NET

Product series code

Communication: LOEU: LoRa EU / NB: NB-IoT

Measuring principles:CO2: $CO_2$ + THP / VOC: VOC + THP

T = temperature H = relative humidity P = pressure

Type of energy supply: BAT: Battery / NET: Power supply unit (old stock)

| Sensor | CO2 | SCW | TH |
|---|---|---|---|
| **$CO_2$** | ✓ | | |
| **temperature** | ✓ | ✓ | ✓ |
| **rel. humidity** | ✓ | ✓ | ✓ |
| **air pressure** | ✓ | | |
| **TAP** | (✓) | (✓) | (✓) |
| **VOC** | (✓) | | |
| **Surface thermometer (e.g. mold prevention)** | | ✓ | |

(✓) optional, ✓ included

# 5. ASSEMBLY AND INSTALLATION

If the sensor is still easily accessible after installation, install the sensor first and activate it after installation!

If the sensor is no longer accessible after installation, activate the sensor first and install it after activation!

Please note:

- Do not insert any objects or body parts into the ventilation slots of the sensor.
- Do not mount the sensor on the ceiling or floor.
- Do not install the sensor at heights above two meters.
- Only install the sensor indoors on a wall in a standard room at a height of 1.50 m to 1.80 m.
- Make sure that the LEDs are on the lower right-hand side when mounting.

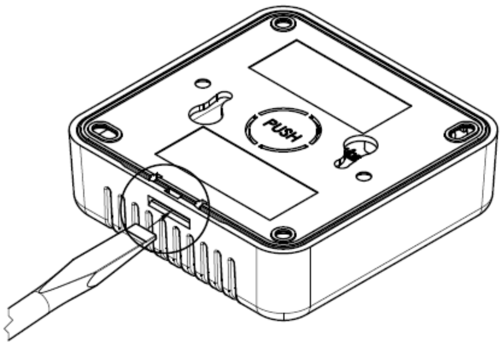
| Mounting type | Description | Recommended accessories | Article code | Included |
|---|---|---|---|---|
| Screw connection | 2x screws | 2x suitable M3 countersunk head screw, wood or sheet metal screw if necessary | FEBR-SCREW-SPAX | Not included |
| Magnets | 2 neodymium pot magnets, adhesive force 16 - 32 kg incl. 2 screws | 2x neodymium magnets (indoor) together 32 kg load capacity | FEBR-MAG-NEO | Not included |
| Gluing | mounting adhesive | Mounting adhesive | FEBR-GLUE | On request / Not included |
| Placing / Laying | / | / | / | / |

Permanent magnets can generate strong magnetic fields, which can be dangerous if handled incorrectly. Therefore, observe the following warnings:

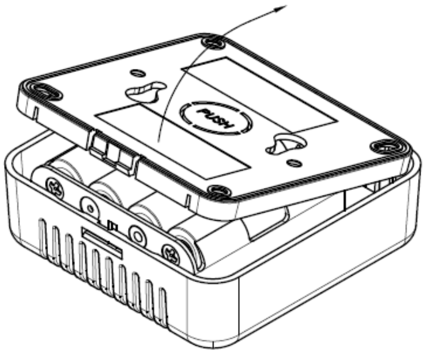
- Protect hands and fingers: Strong magnets can suddenly attract and pinch fingers or skin. This can lead to painful crushing and injuries. Always keep sufficient distance when handling permanent magnets and wear protective gloves if necessary.
- Keep electronic devices away: Magnetic fields can damage electronic devices such as computers, smartphones, credit cards, pacemakers and other sensitive electronics or impair their function. Therefore, always keep a sufficient distance from such devices.

- Be aware of the risk of breakage: Many permanent magnets are made of brittle materials (e.g. neodymium), which can break in the event of sudden impacts or high loads. The splinters can be sharp and cause injuries. Therefore, use the magnets carefully and avoid impacts or excessive loads.
- Health risks: People with pacemakers or other implanted medical devices should avoid contact with strong magnets, as the magnetic fields may interfere with or disable these devices. If necessary, consult a doctor before use.
- Store magnets safely: Keep magnets at a safe distance from each other and from other metal objects. Sudden attraction can lead to damage, injuries or uncontrollable flying around of the objects.
- Danger for children: Permanent magnets are not toys! Small magnets in particular can be life-threatening if swallowed or inhaled and can cause serious internal injuries. Therefore, always keep magnets out of the reach of children.
- Avoid heating: Permanent magnets permanently lose their magnetic force at temperatures above their maximum operating temperature (between 80 and 200 °C, depending on the material). Therefore, do not expose magnets to direct heat or open flames.

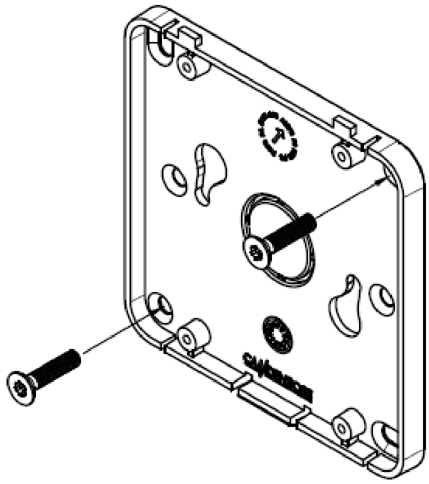
## 5.1 WALL MOUNTING WITH SCREWS



Open the housing by the snap lock and remove the top. Use a screwdriver if necessary.

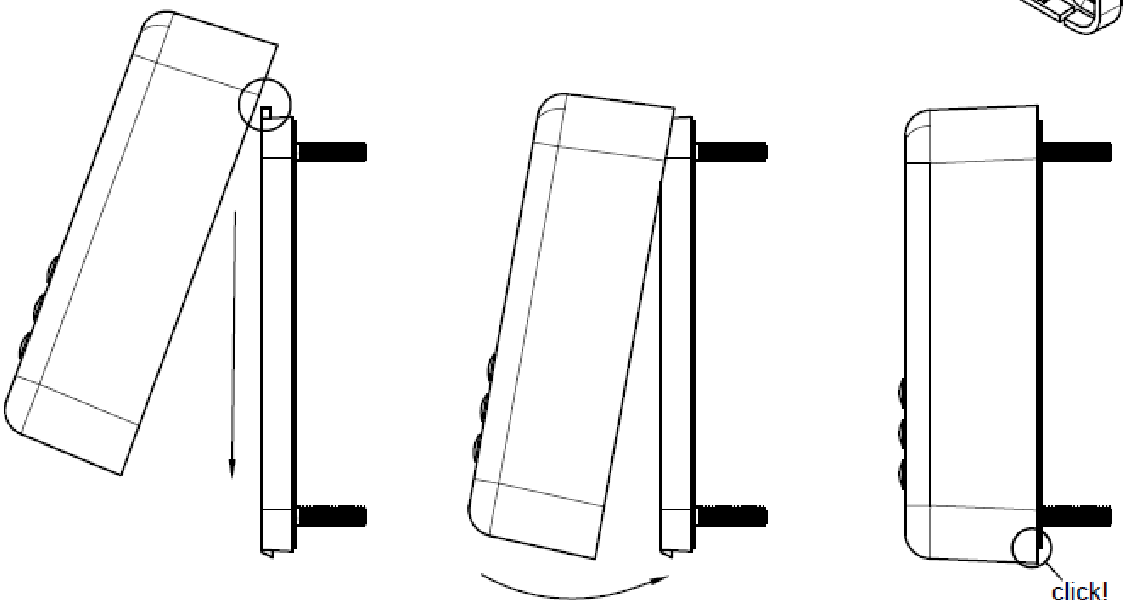


Remove the back of the sensor.



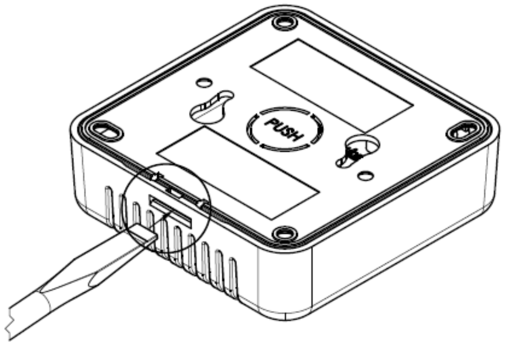
Fasten the back at the desired location using suitable countersunk screws (recommended size M3.5). Alternatively, the two additional eyelets in the corners can also be used. Deviations are possible depending on the surface and mounting method.

Make sure that the suspension is mounted so that the tab for hanging the sensor is at the bottom.

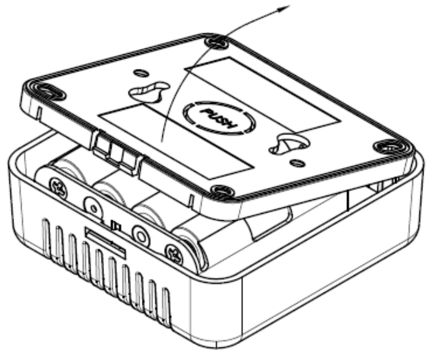



Insert the sensor into the wall bracket from above until it snaps into the upper bracket. Tilt the sensor backwards so that it snaps securely into the holder and the snap lock. Ensure that the sensor is fully seated in the bracket and is firmly attached.

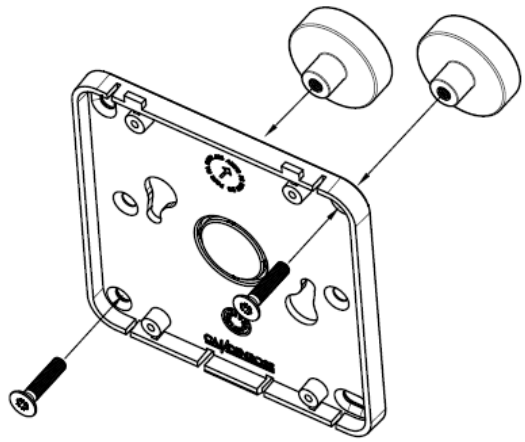
## 5.2 WALL MOUNTING WITH MAGNETS



Open the housing by the snap lock and remove the top. Use a screwdriver if necessary.

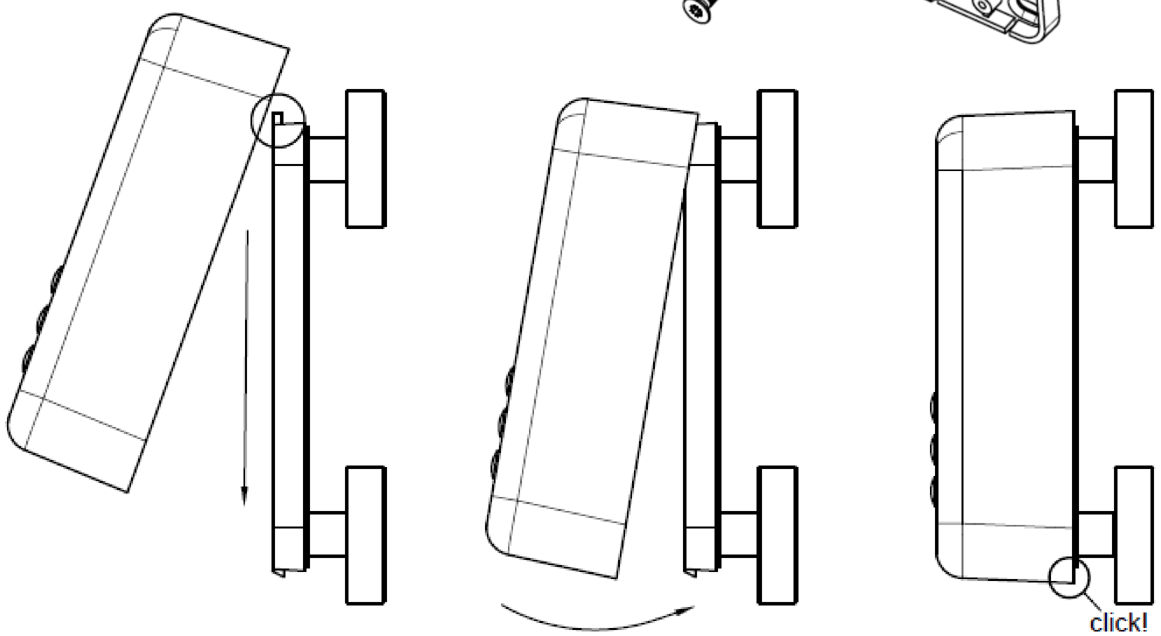


Remove the back of the sensor.



First attach the magnets to the back by screwing them to the round holes in the corners. You can then attach them to the desired position.

Make sure that the suspension is mounted so that the tab for attaching the sensor is at the bottom.

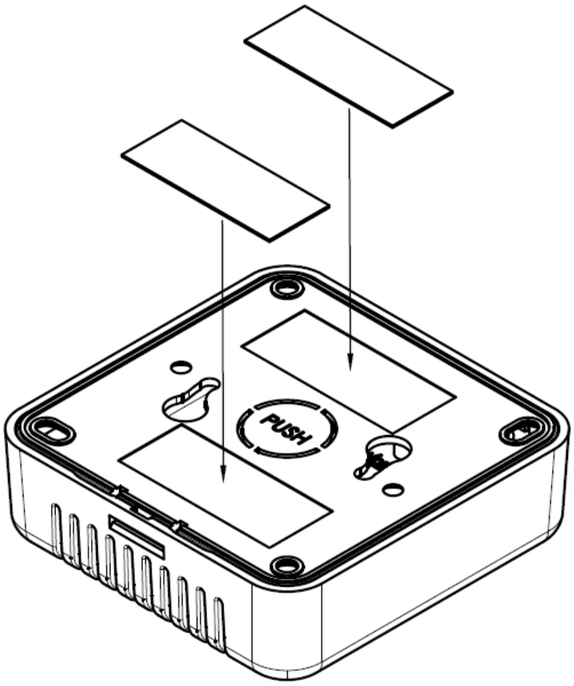



Insert the sensor into the wall bracket from above until it snaps into the upper bracket. Tilt the sensor backwards so that it snaps securely into the holder and the snap lock. Ensure that the sensor is fully seated in the bracket and is firmly attached.

## 5.3 WALL MOUNTING WITH ADHESIVE STRIPS

Make sure that the surface is clean, dry, smooth and adhesive before using this type of installation, as an uneven or dusty surface can impair the adhesion of the adhesive strips. Avoid mounting on rough, porous or damp surfaces, as this can reduce the adhesive strength and cause the mount to peel off. Press the mount firmly for a few seconds after sticking it on to ensure an optimum bond between the adhesive strips and the surface.

Stick the two double-sided adhesive tapes to the back of the sensor in the space provided. The tape should be approximately 41mm x 17mm in size and 1mm deep. When attaching it to the wall, make sure that the snap lock is at the bottom. Once the bracket is attached, check that it is secure by pulling gently on it. Make sure that the adhesive strips adhere completely and that no corners come loose.



# 6. COMMISSIONING AND USE

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Please note that the housing or electronics may be damaged if knives
or other sharp objects are used.
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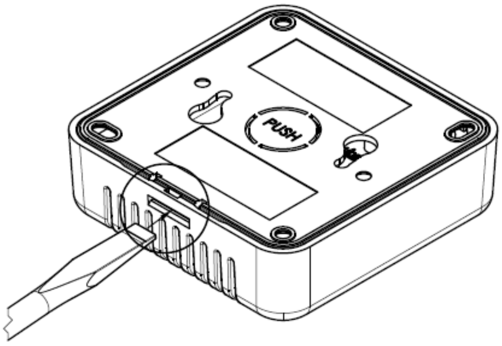
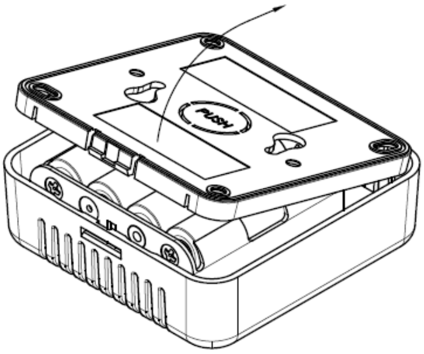
## 6.1 COMMISSIONING THE SENSOR

Open the sensor using the tabs provided (on the bottom edge in the picture). If necessary, carefully use a blunt object (e.g. a screw driver).

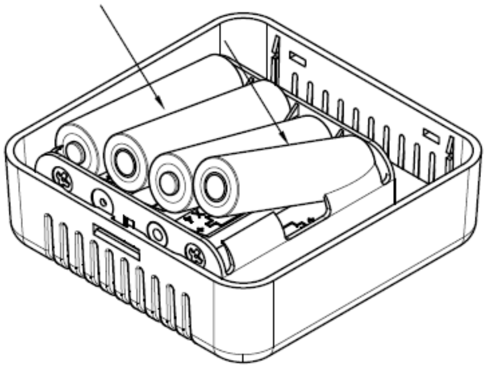



Remove the back of the Sensor

Now insert the cells. To achieve the specified runtimes and the performance specified in the data sheets, the following primary cells are permitted:

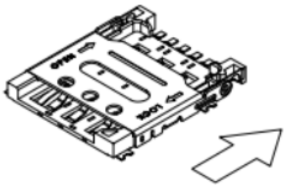
- Energizer® Ultimate Lithium Batteries - AA
- VARTA ULTRA LITHIUM Mignon AA



4 Cells are required per sensor.

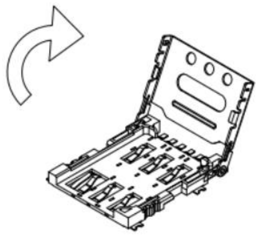
If you have an NB-IoT sensor and you want to insert your own SIM card, insert the SIM card now.

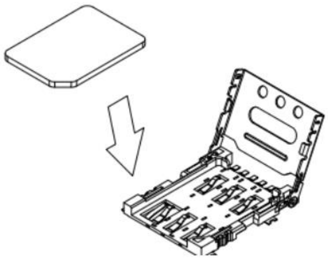
SIM Card holder



Unlock the metal lid by sliding it upwards.

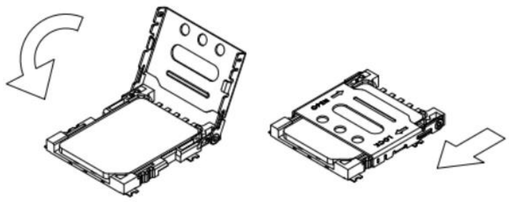
Open it to insert the Nano SIM card



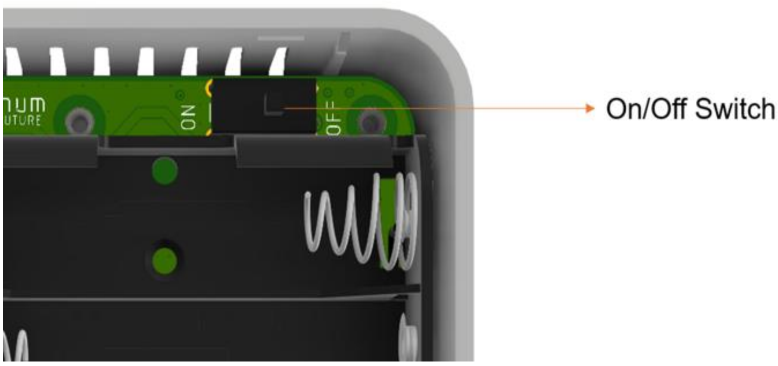


Now place your SIM card in the holder provided.

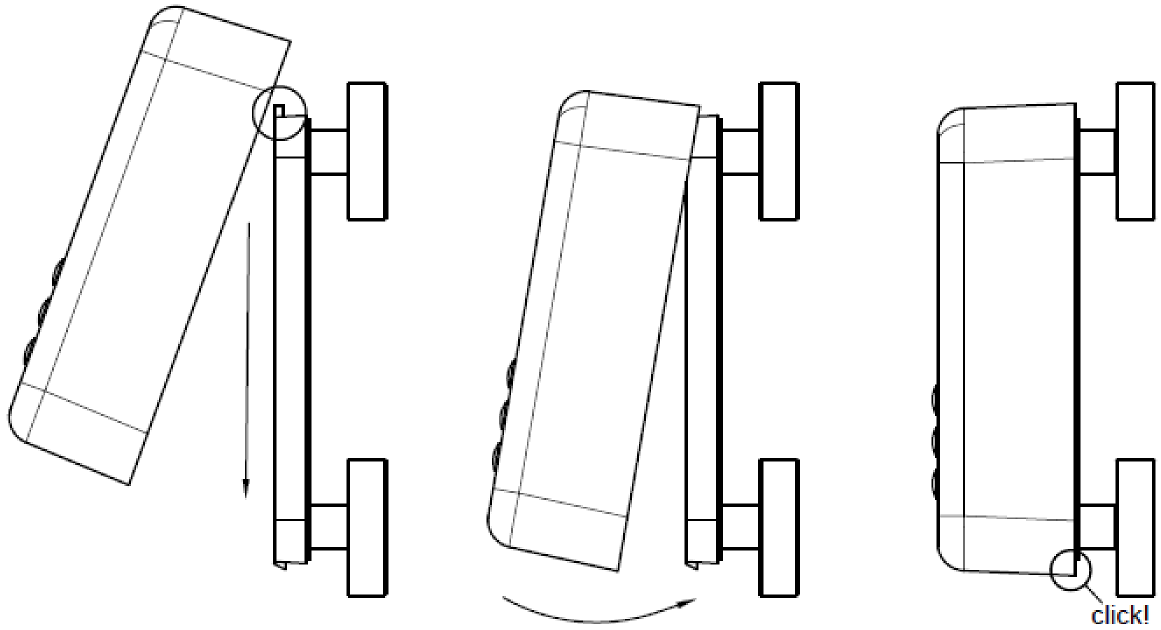
Close the device and lock it again by sliding the metal cover downwards.



Press the switch to put the device into operation. After flipping the switch, the sensor starts, the 3 LEDs on the front flash and a tone sequence can be heard. The sensor then carries out a measurement. The LED traffic light shows you the result.



Insert the sensor into the wall bracket from above until it snaps into the upper bracket. Tilt the sensor backwards so that it snaps securely into the holder and the snap lock. Ensure that the sensor is fully seated in the bracket and is firmly attached.

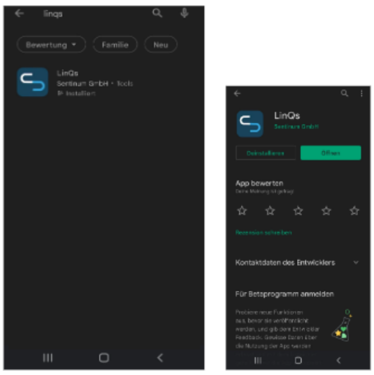



## 6.2 NFC PARAMETRIZATION AND POSITION OF THE TAG

Activation takes place via an NFC app. A smartphone is required for this. The app can be downloaded from the respective app stores. Simply search for "Sentinum LinQs" and download the LinQs app.

Google Play Store

iOS App Store





First locate the tag on the sensor and then the reader on your end device. The position of the NFC tag can be found at the position of the orange arrow.

## 6.3 POSSIBLE APPLICATIONS OF THE SENSORS

The Febris IoT sensors are specially designed for monitoring indoor air quality. They record precise environmental values such as $CO_2$ concentration, temperature and humidity and enable continuous remote monitoring. The recorded data helps to optimize the indoor climate, reduce energy costs and minimize health and structural risks. The sensors are ideal for use in offices, conference rooms, schools, museums, storage rooms, hospitals and private homes.

Areas of application and benefits:

- Air quality monitoring: the $CO_2$ sensor measures the carbon dioxide content in the air and helps to ensure a healthy indoor climate. Excessively high $CO_2$ levels can cause tiredness and concentration
- Mould prevention: The temperature and humidity sensor detects critical humidity values and enables early action to prevent mould. Mold growth can be prevented by automatically controlling ventilation systems or dehumidifiers.
- Increase energy efficiency: The sensors enable heating, ventilation and air conditioning systems to be controlled according to demand. This saves energy, as the climate control is only activated when it is actually necessary.
- Protect sensitive materials: In museums, archives or storage rooms, the continuous monitoring of temperature and humidity ensures that works of art, documents and other sensitive objects remain protected from damage caused by unfavorable climatic conditions.
- Meeting regulatory requirements: In many areas, such as public facilities or the pharmaceutical industry, there are regulations regarding maximum $CO_2$ values or permissible temperature and humidity ranges. The sensors support compliance with these specifications and provide transparent documentation.
- Permanent remote monitoring: all measured values can be retrieved even over long distances, enabling a rapid response to critical changes. Building managers and operators are automatically notified as soon as threshold values are exceeded.

The use of Febris IoT sensors ensures a healthy, efficient and safe indoor climate. The sensors provide a reliable basis for data-based decisions on air quality and climate control.

# 7. COMMUNICATION WITH THE INTERFACE

## 7.1 JOIN & SEND

After activation, the sensor automatically connects to the network. If the first connection attempt is unsuccessful, it switches to an exponential repeat loop. As soon as the connection has been successfully established, the sensor starts taking measurements and sends data according to the configured settings.

If the first connection attempt is successful, a measurement is taken immediately and the first data packet is sent. Otherwise, the sensor starts the repeat loop, but does not perform a measurement or data transmission immediately after each subsequent connection attempt.

The first data packet sent (uplink) usually contains an alarm variable, as the measured value is compared with an internal reference value. The link check settings are deactivated by default.

## 7.2 DUTY-CYCLE-RESTRICTION

The module strictly adheres to the duty cycle restriction, which can lead to delays in transmission. A maximum of six packets can be stored in a queue, which are then sent with a time delay in accordance with the duty cycle rule.

As soon as the queue is full, all additional incoming packets are discarded. New packets are only accepted again when there are fewer than six packets in the queue.

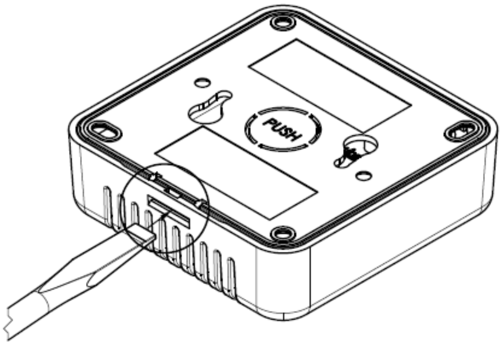
## 8. CARE AND CLEANING

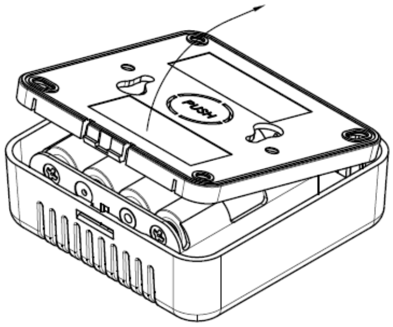
To ensure that the sensor functions reliably and has a long service life, it should be maintained regularly. Observe the following instructions:

- Clean the housing, especially the ventilation slots of the sensor, with a dry or slightly damp microfiber cloth. Make sure that no moisture penetrates the device.
- Carry out cleaning regularly, especially in dusty or pollen-rich environments, to ensure the long-term functionality of the sensor.
- Regularly check that the sensor and the holder are firmly in place.
- Do not use cleaning agents containing alcohol or solvents, as these can damage the surface of the sensor.
- Do not use compressed air or other intensive cleaning methods, as these can damage sensitive sensor components.

## 9. CHANGING THE BATTERY

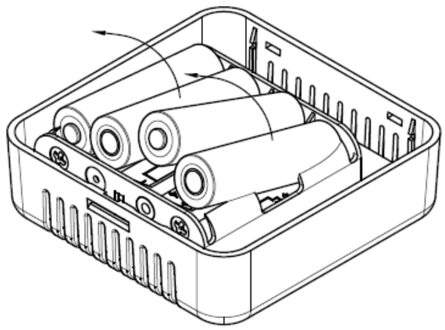
Open the housing by the snap lock and remove the top. Use a screwdriver if necessary.

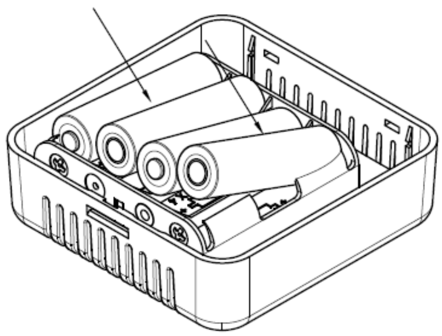




Remove the back of the sensor.

Remove the old batteries.

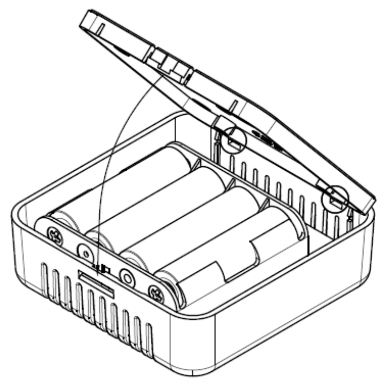


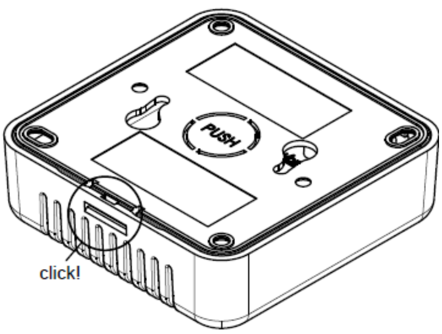


Insert the new cells into the holder provided.

Recommended cells: 4xAA VARTA LONGLIFE Power cells, as supplied. Please note that the cells can affect the runtime of the product depending on the manufacturer, charge and charge status.

Close the cover of the holder again by first inserting the cover into the two recesses and then moving it back down on the other side.






Make sure that the sensor engages securely in the snap lock. Ensure that the sensor is fully seated in the holder and is firmly attached.

# 10. LABELING AND CERTIFICATION